

AI in Context 2026, valueslab.github.io

AI and Mental States

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Plan for today

- “Can AI think?” “Is AI intelligent?”
- It is notoriously hard to explain what thinking is, or what intelligence is.
- Workaround: can AI do what humans do?
- Do AIs have beliefs, and more generally, mental states?
- Ascriptionism, traditionalism, empirical surveys, weak ascriptionism

Belief and Desire/Intention

- When we ask ourselves what's going on in the human mind, we may come up with something like the following:
 - We *believe* things about the world.
 - We *want* things to be the case in the world.

Belief and Desire/Intention



The cat is hunting. It *believes* that there is a mouse over there. It *intends* to catch it.

Ascriptions of Mental States

When we talk like this, we ascribe mental states to the cat.

- “believes”: we ascribe a belief (that there is a mouse over there)
- “intends”: we ascribe intention, that it does what it does in ways that are guided by its desires/wants.

Do Animals Have Beliefs?

Do cats have beliefs? Isn't belief a mental state that is distinctive of human minds?

VOTE 1:

YES (cats have beliefs), NO (cats don't have beliefs)

Analogous Questions About AI

It is contested whether AIs (AI systems, models, LLMs) have beliefs.

- Like animals, AIs don't have human minds.
- But other than cats, AIs are not *natural* creatures; they are *artifacts*.
- Cats *perceive* the world; these perceptions orient how they *move* in the world. In these respects, cats are more like us than AIs.
- On the other hand, AIs can produce answers to questions in linguistic form, which cats can't.

Option 1: Ascriptionism

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- If a behavior is best explained by ascribing beliefs to X, X has beliefs.
- What about the worry that the AI doesn't have a mind?
- Forget about the mind!

Option 1: Ascriptionism—Why?

- It's not clear what it would mean to ascribe beliefs, desires, intentions, etc., to AIs.
- But it's also not entirely clear what it means to ascribe beliefs, desires, intentions, etc., to us!

Option 1: Ascriptionism—Why? (Continued)

- Talk about beliefs, desires, intentions, etc., plays an *explanatory* role.
- We explain behavior by ascribing these states.
- If that's an important dimension of talk about mental states, why not do the same w.r.t. AIs?

Ascriptionism (based on Schwitzgebel 2024)

Interpretationism: A revision of dispositionalism.

Daniel Dennett: When we explain observable behavior, we take three stances: stance of physics, design stance (about functions of organs, etc.), or the intentional stance, where we ascribe intentions and other mental states.

According to interpretationism: “[t]he system has the particular belief that P if its behavior conforms to a pattern that can be effectively captured by taking the intentional stance and attributing the belief that P.” (Schwitzgebel 2024)

Ascriptionism (based on Schwitzgebel 2024)

Dispositionalism: Beliefs are behavioral dispositions. For someone to believe that P is for them to have some behavioral dispositions pertaining to P.

For example, for the cat to believe that there is a mouse over there is to walk a certain way, listen closely, and so on.

Objection 1: This is reductionism. Why reduce beliefs to behavior?

Objection 2: What about beliefs that don't have obvious behavioral correlates?

Option 2: Traditionalism

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- Traditionalist: Beliefs, desires, intentions, etc., are mental/psychological states that are distinctively human.
- Murray Shanahan (2024): It is tempting to say things like “the AI believes,” but it is a form of anthropomorphism.
- AIs simply aren’t the kind of thing that has beliefs. They are “generative mathematical models of the statistical distribution of tokens in the vast public corpus of human-generated text...”

Option 2: Traditionalism's Anthropomorphism Charge

- What is anthropomorphism?
 - Perhaps: the ascription of human states to non-human entities.
- What's wrong with anthropomorphism?
 - Presumably: it involves the mistaken belief that something that doesn't have human-like states has such states.
- Why is anthropomorphism seemingly so pervasive?
 - Evolutionary anthropology and cognitive science of religion: anthropomorphism as cognitive device that augmented human fitness (Damiano and Dumouchel 2018).

Option 2: Traditionalism

- Suppose Shanahan (2024) is right: it makes no sense to ascribe human mental states to AIs.
- Does this beg the question? Sure, AIs don't have human minds. Accordingly, they don't have human mental states.
- But is that all we wanted to know?

Traditionalism vs. Ascriptionism

Whether we are ascriptionists or traditionalists determines our answer to whether AIs have beliefs:

- Ascriptionism: yes
- Traditionalism: no

Minimal Idea 1: Belief as a Propositional Attitude

A couple of examples for propositional attitudes:

1. Sara hopes that there is a class on AI in the Fall.
2. Sara believes that there is a class on AI in the Fall.
3. Sara knows that there is a class on AI in the Fall.

Cognizer—attitude—*that P*.

P [proposition]: *there is a class on AI in the Fall*.

To hope, to believe, to know, and so on, are attitudes to P.

Minimal Idea 2: Truth and Falsity

Belief is a non-factive propositional attitude.

That is, if Sara believes that P, it can either be the case that P (be a fact that P) or not be the case that P (not be a fact that P).

Here's another way of putting this: If Sara believes that P, P can be either true or false.

Minimal Idea 3: Truth as the Aim of Belief

Truth is the *aim* of belief.

When you form a belief, you aim to hold something to be true that really is true.

Cf. Williams (1973)

Do AIs Have Beliefs?

Do AIs have beliefs?

VOTE 2:

Ascriptionism (some version of dispositionalism or interpretationism)

Traditionalism (beliefs are distinctively human mental states that relate to complex attitudes and skills, such as sincerity/insincerity, assertion/lying, etc.)

Intentions



We didn't only talk about beliefs. We also asked whether to ascribe *intentions* to the cat.

Intention, Ends, Responsibility, Etc.

In ascribing intentions to some entity—a human being, a cat, an AI—we talk about it as an *agent*. Here are a few related notions:

- agents
- action
- intention
- ends
- responsibility

Option 3? Empirical Research on Human Attitudes

- So far, we assumed that AIs *either have mental states or not*.
- Empirical work on social robots: “Empirical results show that social robots tend to blur the traditional ontological categories that humans use to describe the world. [...] interactive computational technologies bring people to revise the ontological categories they use to classify objects that, like social robots, are located somewhere in between the terms of the old dichotomies objects that are “sort of alive” or “alive enough” (Turkle, 2011).” (Damiano and Dumouchel 2018)

Option 3?

- What follows from this kind of empirical study?
 - That many people don't quite know whether AI has mental states? That would seem plausible. But it doesn't tell us whether AI has mental states.
- The question of whether AI has mental states isn't merely a matter of theory.
 - Our assessment matters for practical purposes.
 - This becomes evident in the final type of position that we study, where AIs are ascribed responsibility because otherwise there are events in the world that no one is responsible for...

Option 4?

Cibralic and Mattingly (2021) defend a fourth option (next to ascriptionism, traditionalism, and underdetermination).

- In the traditionalist framework, we can't ascribe responsibility to AI, because we can't ascribe representations.
- But we *want* to be able to ascribe responsibility. Why? More on the next slide.
- They propose a minimalist account of representation for AI.

The Motivations of Ascriptionism

Why do we *want* to ascribe representations (and perhaps, more generally, mental states)?

- So far, we discussed something like *inference to the best explanation*: the best way to make sense of a given behavior is to ascribe mental states.
- Cibralic and Mattingly introduce another kind of motivation: we want to be able to distinguish between the responsibility of those who built the AI, and the specific individual outputs, which (in a sense that remains TBD) the AI is responsible for.

Responsibility Gap

Cibralic and Mattingly's motivation responds to the so-called responsibility gap (Andreas Matthias, 2004):

- Those who build an AI are responsible for its overall design.
- But they cannot predict a specific output at a given occasion.
- There is something that someone else should be responsible for.
- That someone else might be the AI.

Does this get us all the way to responsibility?

However...

Why not say that:

- The AI *causes* the outputs.
- In some sense, *no one* is directly responsible for the specific outputs, because AI isn't an agent who is suitably held responsible.
- We can't just assign responsibility to X because we need someone to blame...
- Are the designers of an AI those who, while not directly responsible for the specific outputs, closest to being responsible?

Essay Topic 1

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When we ascribe beliefs, intentions, etc., to AIs, is that just for ease of expression, a *façon de parler*? Do we have philosophical reasons, grounded in what we take AIs to be, to ascribe mental states to them?

Your paper can discuss mental states in general. Alternatively, you can focus on beliefs *or* on desire/intention. You can also focus on yet another mental state/person-like attitude. For example, you can focus on responsibility and the so-called responsibility-gap. If you're interested in language, you can also ask: can an LLM *speak*?

Readings

- Cibralic, Beba and James Mattingly, “Machine agency and representation” (2021)
- Damiano, Luisa and Paul Dumouchel, “Anthropomorphism in Human-Robot Co-Evolution,” *Frontiers in Psychology* (2018)
- Eric Schwitzgebel, “Belief,” The Stanford Encyclopedia of Philosophy (Spring 2024 Edition), Edward N. Zalta & Uri Nodelman (eds.), URL = <<https://plato.stanford.edu/archives/spr2024/entries/belief/>>
- Shanahan, Murray, “Talking about Large Language Models” (2023)
- Turing, Alan M., “Computing Machinery and Intelligence” (1950)
- Williams, Bernard, “Deciding to Believe” (1973)